Power Analysis for Run-In Clinical Trials under AR(1) and General Correlation Structures

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Abstract

Background. The companion Paper 1 of this compendium derived closed-form power formulas for run-in and common close observations under compound symmetry. Compound symmetry assumes that all pairs of observations are equally correlated, regardless of temporal separation. However, this assumption is empirically rejected in most longitudinal-trial data. AR(1), $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k-k'|}$, offers the natural alternative. **Methods.** We derive the variance of the treatment-effect estimator under AR(1) via two routes. The first route applies the Woodbury identity to the AR(1) marginal covariance. The second route uses a moment-based approach based on the cross-covariance between run-in and treatment-period means. The two routes target different estimands: the GLS variance of the slope-difference effect, and the variance of the simple phase-mean difference. The GLS derivation accommodates arbitrary J_0 run-in observations and arbitrary J_2 common close observations. **Results.** We present a taxonomy of five correlation cases and derive change-score correlations in closed form under compound symmetry and AR(1). We compare the two structures numerically, using parameters from Alzheimer's disease neuroimaging trials. When the structures are matched at a common nominal correlation parameter ρ, the AR(1) sample-size estimate is smaller than the compound-symmetric estimate, by 4-36% at $\rho = 0.5$. This deviation grows as ρ increases. However, a narrow anticonservative regime exists at large J_0 and small ρ, where compound symmetry understates the variance by at most about 6%. **Conclusions.** At a matched nominal ρ, the compound-symmetry assumption underlying Paper 1 is generally conservative. That is, it overstates the required sample size, increasingly so as the autocorrelation grows. The exception is a narrow regime with many run-in observations and low autocorrelation, where it is anticonservative by at most about 6%. The AR(1) formulas derived here quantify both effects, and we recommend using them directly when serial correlation is expected. This conclusion is specific to the common-nominal-ρ matching convention stated in the text.

Contents

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1 Introduction

The efficiency of run-in designs depends on the correlation structure of the repeated measurements. We begin with compound symmetry, under which the correlation between any two change scores is constant, and the run-in observations contribute information about the random slope b_i through a simple averaging mechanism. Under AR(1), on the other hand, the correlation decays with temporal separation, so the information content of a run-in observation depends on how far it falls from the treatment period.

Paper 1[1] assumed $R = \sigma^2(I_J + \mathbf{1}_J\mathbf{1}'_J)$, which arises from the change-score parameterization under independent errors.[2] extended the run-in framework to two-period linear mixed models but retained the compound symmetry assumption. Here we extend the framework to AR(1) residual correlation, where $\mathrm{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|}$ for $0 < \rho < 1$.

[3] studied optimal designs for autocorrelated longitudinal data but did not consider the run-in setting.[4] and[5] provide general treatments of correlation structures in longitudinal models.[6] proposed a damped-exponential family of correlation structures with $\mathrm{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|^\theta}$, which contains compound symmetry ($\theta = 0$) and AR(1) ($\theta = 1$) as the two boundary members. The comparison developed here is thus a comparison of the two ends of that family, and the intermediate members offer a route to relaxing the binary choice between them.[7] addressed

the effect of intra-individual variation on change-score analyses, a concern that is amplified under AR(1) correlation.

The `longpower` R package[8] is the CRAN reference implementation for longitudinal-trial power calculations. Its `power.mmrm.ar1()` routine implements the analytic AR(1) sample-size formula of[9] for mixed models for repeated measures, and its slope-based closed-form routines (`edland.linear.power()`, `diggle.linear.power()`) assume random intercepts and slopes with independent or compound-symmetric residuals. None of these routines accommodates run-in or common close observations under a random-slope model, and the closed-form AR(1) derivation developed here supplies that case. As $\rho \to 0$ (no serial correlation), the AR(1) change-score covariance collapses to the compound-symmetric structure of Paper 1, so we see that the present formulas contain the Paper 1 results as a special case.

2 Correlation structures for longitudinal trials

2.1 Taxonomy of cases

We consider five correlation structures for the raw outcomes Y_{ijk}:

1. **Compound symmetry:** $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \rho$ for all $k \neq k'$. This is equivalent to the random intercept model and produces the $R = \sigma^2(I + \mathbf{11}')$ structure in change scores.
2. **AR(1):** $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k'-k|}$. Correlation decays geometrically with lag.
3. **Compound symmetry with variable run-in:** Case 1 with $J_{0,i}$ varying across participants.
4. **AR(1) with variable run-in and common close:** Case 2 with $J_{0,i}$ and $J_{2,i}$ varying.
5. **General (unstructured):** $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \sigma_{|k-k'|}/\sigma^2$.

Throughout, the design quantities J_0, J_1, and J_2 correspond to the arguments `r`, `k`, and `f` of the implementing `runinpower` functions.

2.2 Change-score correlations under each structure

Under the consecutive-difference parameterization ($C_j = Y_j - Y_{j-1}$, successive visits separated by t), the correlation between change scores depends on the raw-outcome correlation structure. Here $a = \sigma^2$ and $b = \sigma_b^2$. We note that the GLS machinery of the next section uses change scores from baseline rather than consecutive differences; the two parameterizations are related by a nonsingular linear map, so the GLS variance is unaffected.

For the run-in design with $J_0 = 2$, $J_1 = 1$, with visits at times $-2t, -t, 0, t$ and consecutive differences C_1, C_2, C_3:

Compound symmetry (independent raw errors; the shared intercept component

cancels in differencing):

$$\mathrm{Corr}(C_1, C_3) = \frac{bt^2}{2a + bt^2} \tag{1}$$

$$\mathrm{Corr}(C_1, C_2) = \frac{-a + bt^2}{2a + bt^2} \tag{2}$$

AR(1): for consecutive differences at lag $m = |j - l| \geq 1$,

$$\mathrm{Var}(C_j) = bt^2 + 2a(1 - \rho) \tag{3}$$

$$\mathrm{Cov}(C_j, C_l) = bt^2 - a\,\rho^{m-1}(1 - \rho)^2 \tag{4}$$

so that

$$\mathrm{Corr}(C_1, C_3) = \frac{bt^2 - a\rho(1 - \rho)^2}{bt^2 + 2a(1 - \rho)} \tag{5}$$

$$\mathrm{Corr}(C_1, C_2) = \frac{bt^2 - a(1 - \rho)^2}{bt^2 + 2a(1 - \rho)} \tag{6}$$

These expressions hold for arbitrary (J_0, J_1, J_2) because they depend only on the lag m. At $\rho = 0$ they reduce, as expected, to the compound-symmetry formulas above.

3 Variance under AR(1): GLS approach

3.1 Marginal covariance

Under AR(1) with residual variance σ^2 and autocorrelation ρ, the raw-outcome covariance is $\mathrm{Cov}(e_{ij}, e_{ij'}) = \sigma^2 \rho^{|j-j'|}$. The change-score residual covariance becomes:

$$R_{jl} = \mathrm{Cov}(e_{ij} - e_{i0}, e_{il} - e_{i0}) = \sigma^2(\rho^{|j-l|} - \rho^{|j|} - \rho^{|l|} + 1) \tag{7}$$

for $j, l \neq 0$.

This R no longer has the rank-one-plus-identity structure, so the Sherman-Morrison formula does not apply. The Woodbury identity, however, still reduces Σ^{-1} to a scalar inversion, because G remains scalar.

3.2 Woodbury computation

The computation proceeds identically to Paper 1:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \tag{8}$$

with $H = (G^{-1} + Z'R^{-1}Z)^{-1}$ still scalar. The difference from Paper 1 is that R^{-1} must now be computed numerically (or via the AR(1) tridiagonal inverse) rather than by Sherman-Morrison.

Table 1: Per-pair $\mathrm{Var}(\hat{\gamma})$ under AR(1) for varying ρ ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	$\rho = 0$	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$
0	3.0000	2.9100	2.7500	2.5100	2.1900
2	2.1569	2.0132	1.6662	1.1157	0.3963
4	1.3268	1.3902	1.2218	0.8508	0.3036

4 Variance under AR(1): moment-based approach

4.1 Cross-covariance between phase means

An alternative to the GLS approach computes the variance of the pre-post test statistic directly from the moments of the phase means. This route follows the summary-measures approach of[10], specialized here to AR(1) correlation. Let $\bar{X}_0 = \frac{1}{J_0}\sum_{i=1}^{J_0} Y_{i,\text{run-in}}$ and $\bar{X}_K = \frac{1}{J_1}\sum_{j=1}^{J_1} Y_{j,\text{trt}}$ denote the run-in and treatment-period means.

Under AR(1) with parameter $c = \rho$, the cross-covariance is:

$$\mathrm{Cov}(\bar{X}_0, \bar{X}_K) = \frac{\sigma^2}{J_0 J_1}\,\rho\, c^{K-1}\,\frac{(1-c^{J_0})(1-c^{J_1})}{(1-c)^2} \tag{9}$$

where $c = \rho$. We note that the symbol c here denotes the autocorrelation and is unrelated to the reparameterization $c = \sigma^2 - \sigma_b^2 t^2$ used in Paper 1. K is the time-index lag from the last run-in visit to the first treatment visit on the visit grid $d = (-J_0, \ldots, -1, 1, \ldots, J_1)$. Because baseline (time 0) is not itself a phase observation, adjacent phases have $K = 2$. The division by $J_0 J_1$ reflects that $\bar{X}_0$ and $\bar{X}_K$ are means of J_0 and J_1 observations respectively. This matches the implementation in `ar1_cov_phase_means()` below.

4.2 Variance of the change score

The variance of the pre-post difference is:

$$\mathrm{Var}(X_K - X_0) = \mathrm{Var}(X_K) + \mathrm{Var}(X_0) - 2\,\mathrm{Cov}(X_K, X_0) \tag{10}$$

The variance of each phase mean involves the sum $S_n = \sum_{i=1}^{n-1}(n-i)c^{i-1}$, which has the closed form:

$$S_n = \frac{n(1-c) - (1-c^n)}{(1-c)^2} \tag{11}$$

```
## Moment-based vs GLS variance (r=2, k=2, rho=0.5):
##   Moment-based phase-mean difference: 1.2188
##   GLS slope difference (sigma_b2=0):  0.6364
##   GLS slope difference (sigma_b2=1):  1.6662
##   (Different estimands; see next section.)
```

Table 2: Sample size per group (MIRIAD parameters, $\Delta = 0.25$, 90% power) under compound symmetry vs AR(1).

J_0	J_1	J_2	N (CS)	N (AR(1), $\rho = 0.5$)
0	2	0	385	371
1	2	0	247	157
2	2	0	145	107
3	2	0	100	85
4	2	0	78	72

5 Relationship between the two approaches

We note at the outset that the GLS approach (Section 3) and the moment-based approach (Section 4) compute different quantities. The GLS variance is the minimum-variance bound under the assumed model, whereas the moment-based variance is the variance of the simple averaged change-score estimator $\bar{d}_1 - \bar{d}_0$.

It is worth emphasizing that the two approaches do not coincide even when $\sigma_b^2 = 0$. At that boundary the GLS estimator still weights the change scores through the AR(1) residual covariance and estimates the slope-difference parameter γ, whereas the moment-based quantity is the variance of the unweighted phase-mean difference $\bar{X}_K - \bar{X}_0$. With $J_0 = 2$, $J_1 = 2$, $\rho = 0.5$, $\sigma^2 = 1$, and $\sigma_b^2 = 0$, the GLS variance is 0.636 while the moment-based variance is 1.219. When $\sigma_b^2 > 0$ the GLS estimator additionally exploits the random-slope structure of $\Sigma = R + ZGZ'$.

The practical value of the moment-based computation is that it requires no specification of σ_b^2 and gives the exact variance of the simple pre-post difference that an analyst might report without fitting a mixed model. That said, it is not a bound on the GLS variance: at the parameters above, the moment-based value (1.219) lies between the GLS variance at $\sigma_b^2 = 0$ (0.636) and at $\sigma_b^2 = 1$ (1.666). Because the two quantities are variances of different estimands, their ratio describes the cost of abandoning the mixed model rather than an efficiency ordering within it. We take up the relative-efficiency question in Paper 4.

6 Robustness to correlation misspecification

The practical question we address here is whether a trial designed under compound symmetry loses substantial efficiency when the true correlation is AR(1), and vice versa. Throughout this section the two structures are matched at a common nominal correlation parameter ρ; we note that the comparison, and its direction, depend on this matching convention.

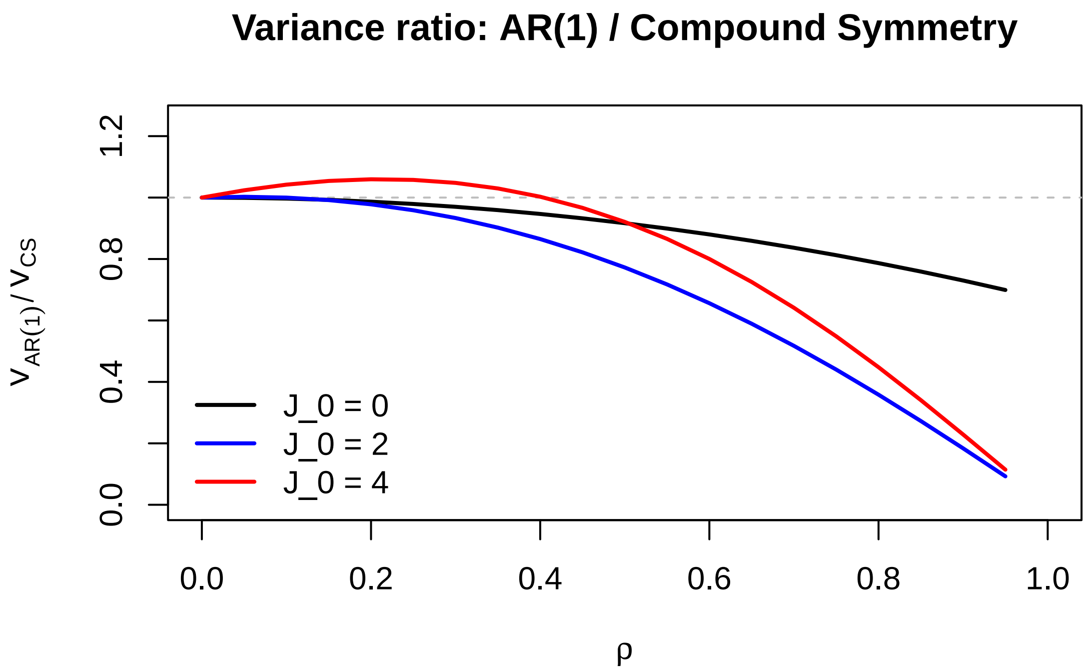

Figure 1: Ratio of AR(1) to compound-symmetry variance at a matched nominal ρ for three run-in configurations ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$). The ratio remains near or below one except at $J_0 = 4$ and small ρ, where it peaks near 1.06.

7 Numerical examples

7.1 Comparison across correlation structures

At the matched nominal $\rho = 0.5$, we see that the AR(1) sample sizes are smaller than the compound-symmetric ones at every J_0, by 3.6% ($J_0 = 0$) to 36.4% ($J_0 = 1$). Positive serial correlation reduces the residual variance of each change score ($R_{jj} = 2\sigma^2(1 - \rho^{|j|}) < 2\sigma^2$), so the compound-symmetric calculation, which corresponds to independent raw errors, is conservative here.

7.2 Sensitivity to rho

8 Discussion

The central practical question is when the compound-symmetry formulas of Paper 1 can be used in place of the AR(1) formulas derived here. The numerical comparisons indicate that the answer is governed by the same slope-to-noise ratio $\sigma_b^2 t^2/\sigma^2$ that governs the value of run-in observations themselves. When this ratio is large, the random slope term ZGZ' dominates the change-score covariance $\Sigma = R + ZGZ'$, the residual term R contributes little, and the choice between an AR(1) and a compound-symmetric R is largely immaterial. When the ratio is small, on the other hand, the residual term dominates and the correlation structure becomes the leading determinant of efficiency.

The direction of the error matters for trial design. When the two structures are matched

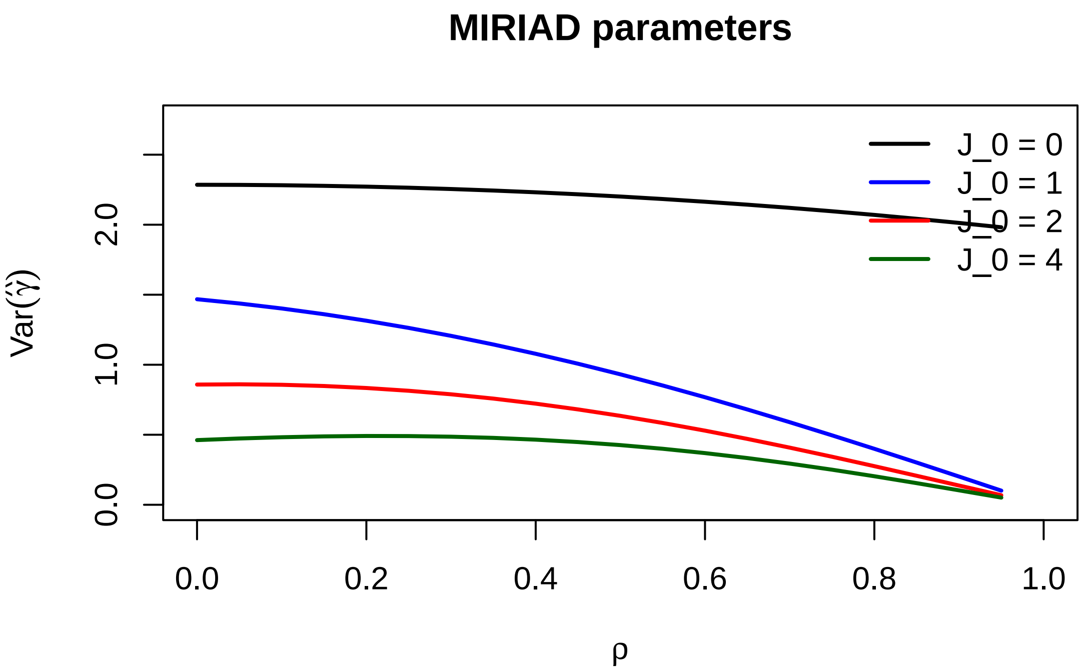

Figure 2: Per-pair variance as a function of ρ under AR(1) for MIRIAD parameters ($\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, $J_1 = 2$, $J_2 = 0$).

at a common nominal ρ, positive serial correlation reduces the residual variance of every change score ($R_{jj} = 2\sigma^2(1-\rho^{|j|})$), so the AR(1) variance lies below the compound-symmetric value for most configurations, and the compound-symmetric sample size is conservative. In the MIRIAD example at $\rho = 0.5$ the AR(1) sample sizes are 4 to 36 percent smaller than the compound-symmetric ones, and the gap widens as ρ grows. The exception is a narrow regime with many run-in observations and low autocorrelation ($J_0 = 4$, $\rho \approx 0.2$), where correlation decay makes distant run-in observations less informative than compound symmetry credits them to be. There the compound-symmetric formula understates the variance, by at most about 6 percent in the configurations examined. We emphasize that both statements are specific to the common-nominal-ρ matching convention; matching the structures on, for example, a common lag-one change-score correlation would alter the comparison.

For applied use, we recommend the following. First, at a matched nominal ρ, the compound-symmetry formula of Paper 1 is conservative for most designs and can serve as a convenient default; the AR(1) formula then refines the sample size downward. Second, when the design places many run-in observations before randomization and the expected autocorrelation is low, the AR(1) formula should be used directly, because the compound-symmetric formula can understate the variance by up to about 6 percent in that regime. Third, when the correlation structure cannot be pinned down from prior data, evaluating the AR(1) formula over a plausible range of ρ and taking the largest resulting variance yields a safe sample size. Finally, we note that the moment-based computation of Section 4 addresses the simple pre-post estimator and is not a bound on the GLS variance.

Scope: balanced designs only. All results above assume that every participant contributes the same number of run-in (J_0) and common close (J_2) observations. When participants differ in $J_{0,i}$ or $J_{2,i}$ (for example, staggered enrollment with a fixed calendar close date), the total variance is a weighted sum across strata, $n^{-2} \sum_{s=1}^{S} n_s \ \mathrm{Var}(X_K - X_0)|_{J_0=J_{0,s},\, J_2=J_{2,s}}$, evaluated stratum-by-stratum with the compound-symmetry or AR(1) formulas derived above and combined by the delta method. This stratified extension is straightforward in principle for the compound-symmetry route, since each stratum uses the moment-based formula of Section 4 directly, but it has not been derived or numerically verified here for either correlation structure, and we do not claim it in the results above. Unbalanced-visit designs are left to future work.

9 Conflict of interest

The author declares no conflict of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

The `runinpower` R package implementing the AR(1) variance formulas (`covariance-ar1.R`), the validation tests, and the shared bibliography across the four-paper compendium are openly available in the project repository at https://github.com/rgt47/runinpower. Specific paths are listed in the Reproducibility section below.

12 Reproducibility

This manuscript was rendered with R 4.4.x. The principal packages used are the in-package `runinpower` source (the AR(1) machinery is in `covariance-ar1.R`), `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. The AR(1) variance formulas are deterministic given fixed input parameters. No random-number generation is used anywhere in this manuscript, so re-running the code chunks reproduces every table and figure exactly.

Data and code availability. The R package source is at `R/`. The companion manuscripts (Papers 1, 3, 4) are at `analysis/report/{01,03,04}-*/`. The manuscript source and the shared bibliography (`references.bib`, a symlink to `../../references.bib`) are at `analysis/report/02-correlation/`.

13 References

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